

Linux Practical Examination

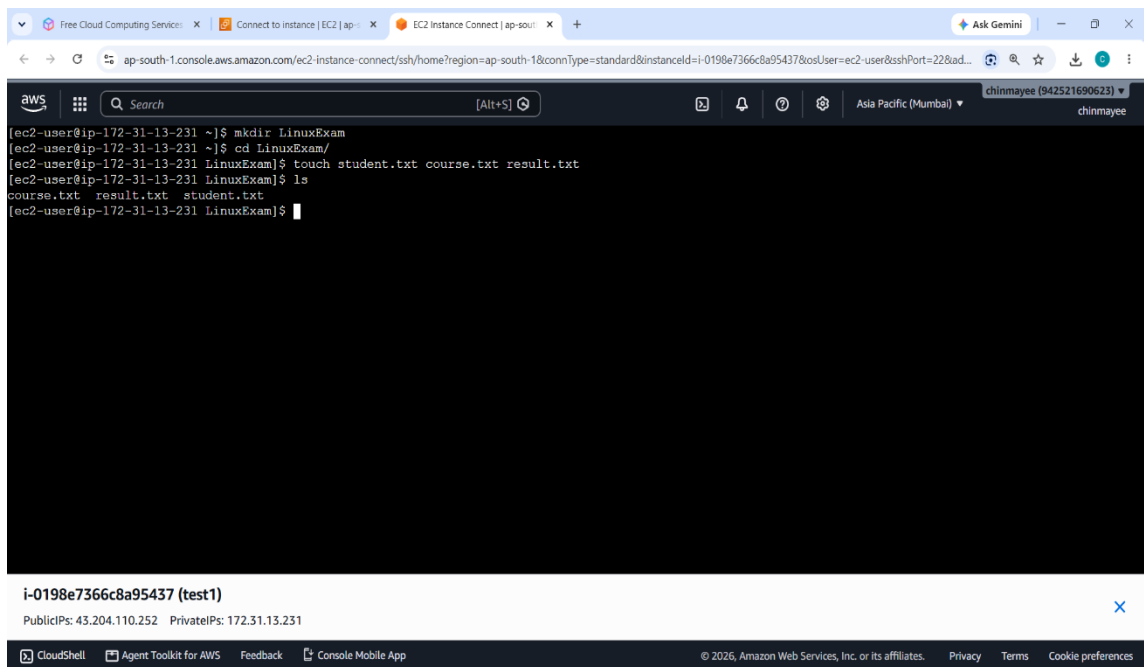
Section A — File & Directory Management

Q1. Basic File Operations ☐ Create

directory named LinuxExam.

- Navigate into the directory.
- Create: o student.txt o course.txt o result.txt
- Display the current working directory.
- Display the contents of the directory.

Ans :



```
[ec2-user@ip-172-31-13-231 ~]$ mkdir LinuxExam
[ec2-user@ip-172-31-13-231 ~]$ cd LinuxExam/
[ec2-user@ip-172-31-13-231 LinuxExam]$ touch student.txt course.txt result.txt
[ec2-user@ip-172-31-13-231 LinuxExam]$ ls
course.txt  result.txt  student.txt
[ec2-user@ip-172-31-13-231 LinuxExam]$
```

i-0198e7366c8a95437 (test1)
PublicIPs: 43.204.110.252 PrivateIPs: 172.31.13.231

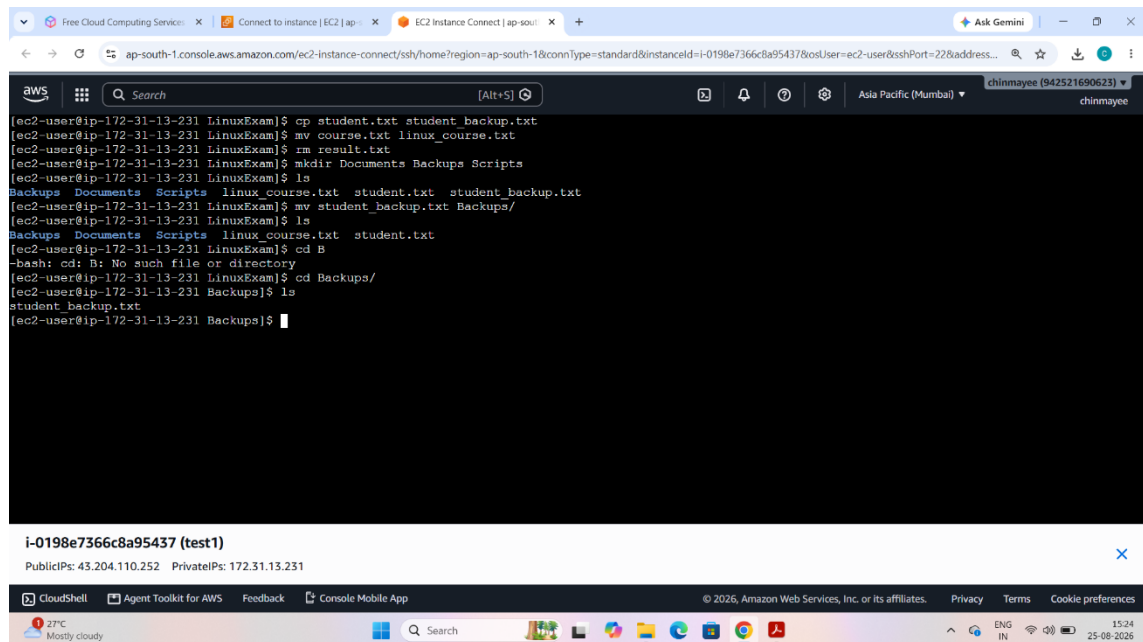
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Q2. File Management

Inside LinuxExam:

- Copy student.txt as student_backup.txt.
- Rename course.txt to linux_course.txt.
- Delete result.txt.
- Create directories: o Documents o Backups o Scripts
- Move student_backup.txt into Backups.
- Display the directory structure.

Ans:



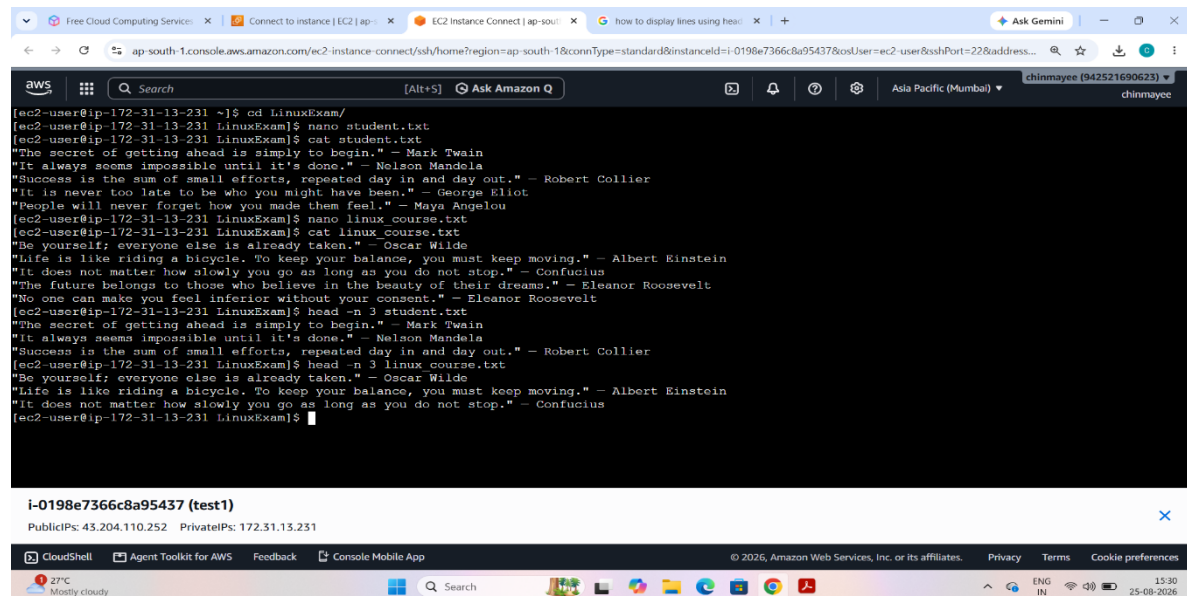
```
[ec2-user@ip-172-31-13-231 LinuxExam]$ cp student.txt student_backup.txt
[ec2-user@ip-172-31-13-231 LinuxExam]$ mv course.txt linux_course.txt
[ec2-user@ip-172-31-13-231 LinuxExam]$ rm result.txt
[ec2-user@ip-172-31-13-231 LinuxExam]$ mkdir Documents Backups Scripts
[ec2-user@ip-172-31-13-231 LinuxExam]$ ls
Backups Documents Scripts linux_course.txt student.txt student_backup.txt
[ec2-user@ip-172-31-13-231 LinuxExam]$ mv student_backup.txt Backups/
[ec2-user@ip-172-31-13-231 LinuxExam]$ ls
Backups Documents Scripts linux_course.txt student.txt
[ec2-user@ip-172-31-13-231 LinuxExam]$ cd B
-bash: cd: B: No such file or directory
[ec2-user@ip-172-31-13-231 LinuxExam]$ cd Backups/
[ec2-user@ip-172-31-13-231 Backups]$ ls
student_backup.txt
[ec2-user@ip-172-31-13-231 Backups]$
```

Q3. File Content Operations ☐ Add at least 5 lines of student

information to student.txt.

- Add at least 5 lines of course information to linux_course.txt. ☐ Display the contents using cat.
- Display the first 3 lines using head.
- Display the last 2 lines using tail.
- Count the number of lines, words, and characters in student.txt

Ans:



```
[ec2-user@ip-172-31-13-231 ~]$ cd LinuxExam/
[ec2-user@ip-172-31-13-231 LinuxExam]$ nano student.txt
[ec2-user@ip-172-31-13-231 LinuxExam]$ cat student.txt
"The secret of getting ahead is simply to begin." - Mark Twain
"It always seems impossible until it's done." - Nelson Mandela
"Success is the sum of small efforts, repeated day in and day out." - Robert Collier
"It is never too late to be who you might have been." - George Eliot
"People will never forget how you made them feel." - Maya Angelou
[ec2-user@ip-172-31-13-231 LinuxExam]$ nano linux_course.txt
[ec2-user@ip-172-31-13-231 LinuxExam]$ cat linux_course.txt
"Be yourself; everyone else is already taken." - Oscar Wilde
"Life is like riding a bicycle. To keep your balance, you must keep moving." - Albert Einstein
"It does not matter how slowly you go as long as you do not stop." - Confucius
"The future belongs to those who believe in the beauty of their dreams." - Eleanor Roosevelt
"No one can make you feel inferior without your consent." - Eleanor Roosevelt
[ec2-user@ip-172-31-13-231 LinuxExam]$ head -n 3 student.txt
"The secret of getting ahead is simply to begin." - Mark Twain
"It always seems impossible until it's done." - Nelson Mandela
"Success is the sum of small efforts, repeated day in and day out." - Robert Collier
[ec2-user@ip-172-31-13-231 LinuxExam]$ head -n 3 linux_course.txt
"Be yourself; everyone else is already taken." - Oscar Wilde
"Life is like riding a bicycle. To keep your balance, you must keep moving." - Albert Einstein
"It does not matter how slowly you go as long as you do not stop." - Confucius
[ec2-user@ip-172-31-13-231 LinuxExam]$
```

The screenshot shows a terminal window within the AWS CloudShell interface. The terminal prompt is `[ec2-user@ip-172-31-13-231 LinuxExam]$`. The user has executed the following commands:

- `tail -n 2 student.txt`: Displays the last two lines of `student.txt`, which are:
"It is never too late to be who you might have been." - George Eliot
"People will never forget how you made them feel." - Maya Angelou
- `tail -n 2 linux_course.txt`: Displays the last two lines of `linux_course.txt`, which are:
"The future belongs to those who believe in the beauty of their dreams." - Eleanor Roosevelt
"No one can make you feel inferior without your consent." - Eleanor Roosevelt
- `wc student.txt`: Shows the word count for `student.txt`, which is 5 lines, 65 words, and 356 bytes.

The terminal output is as follows:

```
[ec2-user@ip-172-31-13-231 LinuxExam]$ tail -n 2 student.txt
"It is never too late to be who you might have been." - George Eliot
"People will never forget how you made them feel." - Maya Angelou
[ec2-user@ip-172-31-13-231 LinuxExam]$ tail -n 2 linux_course.txt
"The future belongs to those who believe in the beauty of their dreams." - Eleanor Roosevelt
"No one can make you feel inferior without your consent." - Eleanor Roosevelt
[ec2-user@ip-172-31-13-231 LinuxExam]$ wc student.txt
 5  65 356 student.txt
[ec2-user@ip-172-31-13-231 LinuxExam]$
```

Below the terminal window, the instance details for `i-0198e7366c8a95437 (test1)` are shown, including PublicIPs (43.204.110.252) and PrivateIPs (172.31.13.231).

Section B — Users, Groups & Permissions

Q4. User Management

Create a user named `student01`.

- Set a password.
- Verify that the user exists.
- Display the user's UID and GID.
- Display the user's home directory.
- Switch to the `student01` account and verify the current username.

Ans:

The screenshot shows a terminal window within the AWS CloudShell interface. The user has executed the following commands:

- `sudo useradd student01`: Creates the user `student01`.
- `sudo passwd student01`: Sets a password for `student01`. The prompt asks for a new password and its confirmation. The password is set successfully.
- `staprunpriv student01`: Switches to the `student01` user.
- `id student01`: Displays the user's identity information: `uid=1001(student01) gid=1001(student01) groups=1001(student01)`.
- `su - student01`: Switches to the `student01` user.
- `pwd`: Displays the current directory: `/home/student01/`.
- `ls`: Lists the contents of the current directory, showing `-bash: /home/student01/: is a directory`.

The terminal output is as follows:

```
[ec2-user@ip-172-31-13-231 ~]$ sudo useradd student01
[ec2-user@ip-172-31-13-231 ~]$ sudo passwd student01
Changing password for user student01.
New password:
Retype new password:
Sorry, passwords do not match.
New password:
Retype new password:
passwd: all authentication tokens updated successfully.
[ec2-user@ip-172-31-13-231 ~]$ id st
staprunpriv student01
[ec2-user@ip-172-31-13-231 ~]$ id student01
uid=1001(student01) gid=1001(student01) groups=1001(student01)
[ec2-user@ip-172-31-13-231 ~]$ su - student01
Password:
[student01@ip-172-31-13-231 ~]$ /home/student01/
-bash: /home/student01/: is a directory
[student01@ip-172-31-13-231 ~]$
```

Below the terminal window, the instance details for `i-0198e7366c8a95437 (test1)` are shown, including PublicIPs (13.252.206.60) and PrivateIPs (172.31.13.231).

Q6. File Permissions

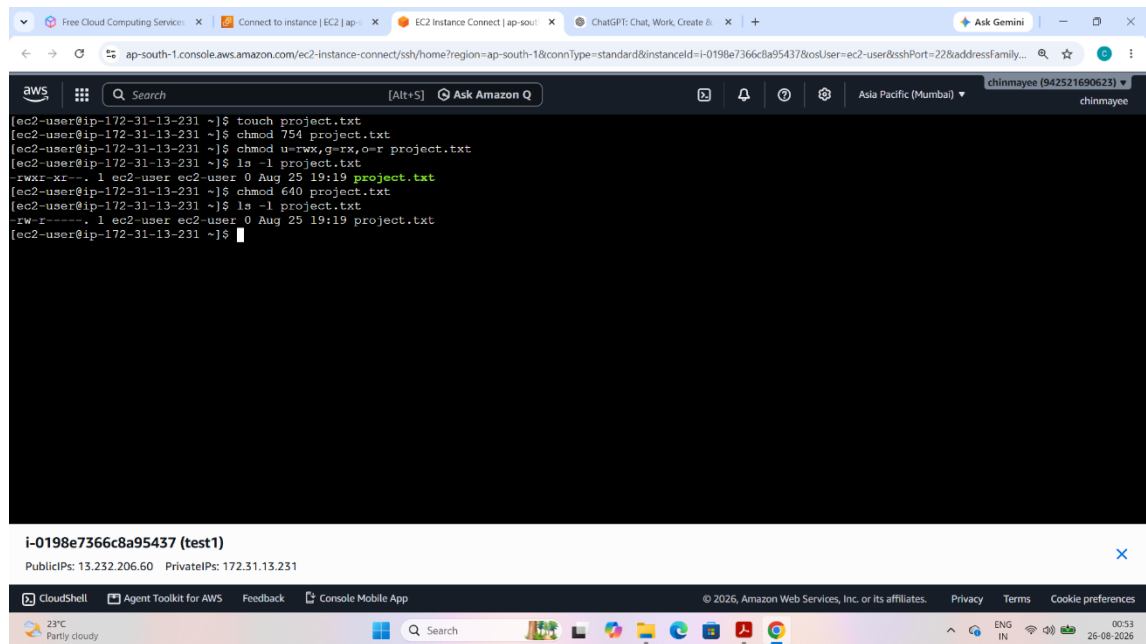
Create project.txt and:

- Give owner Read, Write and Execute permission.
- Give Group Read and Execute permission.
- Give Others Read permission only.
- Verify the permissions.

Expected: 754

- Change the permission to 640.

Ans:



```
aws
[ec2-user@ip-172-31-13-231 ~]$ touch project.txt
[ec2-user@ip-172-31-13-231 ~]$ chmod 754 project.txt
[ec2-user@ip-172-31-13-231 ~]$ chmod u=rwx,g=rx,o=r project.txt
[ec2-user@ip-172-31-13-231 ~]$ ls -l project.txt
-rwxr-xr--. 1 ec2-user ec2-user 0 Aug 25 19:19 project.txt
[ec2-user@ip-172-31-13-231 ~]$ chmod 640 project.txt
[ec2-user@ip-172-31-13-231 ~]$ ls -l project.txt
-rw-r-----. 1 ec2-user ec2-user 0 Aug 25 19:19 project.txt
[ec2-user@ip-172-31-13-231 ~]$
```

i-0198e7366c8a95437 (test1)
PublicIPs: 13.232.206.60 PrivateIPs: 172.31.13.231

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Q7. Permission Challenge

Create:

LinuxExam/

```
├── public/
├── private/
└── shared/
```

Configure permissions so that:

- public can be accessed by everyone.
- private can be accessed only by its owner.
- shared can be accessed by the owner and group members. ☐ Verify the permissions using `ls -ld`.

The screenshot displays the AWS Management Console interface for an EC2 Instance Connect session. The terminal window shows the following output:

```

aws
[Search] [Alt+S]
Asia Pacific (Mumbai) chinmayee (942321690623) chinmayee

Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

Last login: Tue Aug 25 18:51:40 2026 from 13.233.177.4
[ec2-user@ip-172-31-13-231 ~]$ mkdir -p LinuxExam/{public,private,shared}
[ec2-user@ip-172-31-13-231 ~]$ chmod 777 LinuxExam/public/
[ec2-user@ip-172-31-13-231 ~]$ chmod 700 LinuxExam/private/
[ec2-user@ip-172-31-13-231 ~]$ chmod 770 LinuxExam/shared/
[ec2-user@ip-172-31-13-231 ~]$ ls -ld LinuxExam/{public,private,shared}
drwx-----. 2 ec2-user ec2-user 6 Aug 25 19:41 LinuxExam/private
drwxrwxrwx. 2 ec2-user ec2-user 6 Aug 25 19:41 LinuxExam/public
drwxrwx---. 2 ec2-user ec2-user 6 Aug 25 19:41 LinuxExam/shared
[ec2-user@ip-172-31-13-231 ~]$
  
```

Below the terminal window, the instance ID **i-0198e7366c8a95437 (test1)** is displayed, along with its Public IP (13.232.206.60) and Private IP (172.31.13.231).

Q8. Package Management

- Ans:

```
Free Cloud Computing Services | Connect to instance | EC2 | ap-south-1 | EC2 Instance Connect - ap-south-1 | +
ap-south-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=ap-south-1&connType=standard&instanceId=i-0198e7366c8a95437&osUser=ec2-user&sshPort=22&address...
AWS Search [Alt+S] Ask Amazon Q Asia Pacific (Mumbai) chinmayee (942521690623) chinmayee

[ec2-user@ip-172-31-13-231 ~]$ sudo dnf update -y

Amazon Linux 2023 Kernel Livepatch repository
Dependencies resolved.
Nothing to do.
Complete!

[ec2-user@ip-172-31-13-231 ~]$ sudo yum install httpd -y
Last metadata expiration check: 0:01:03 ago on Tue Aug 25 10:22:47 2026.
Dependencies resolved.

Package Architecture Version Repository Size
Installing:
httpd x86_64 2.4.68-1.amzn2023.0.1 amazonlinux 46 k
Installing dependencies:
apr x86_64 1.7.5-1.amzn2023.0.4 amazonlinux 129 k
apr-util x86_64 1.6.3-1.amzn2023.0.2 amazonlinux 97 k
apr-util-ldap x86_64 1.6.3-1.amzn2023.0.2 amazonlinux 13 k
generic-logos-httpd noarch 18.0.0-12.amzn2023.0.3 amazonlinux 19 k
httpd-core x86_64 2.4.68-1.amzn2023.0.1 amazonlinux 1.4 M
httpdfilesystem noarch 2.4.68-1.amzn2023.0.1 amazonlinux 12 k
httpd-tools x86_64 2.4.68-1.amzn2023.0.1 amazonlinux 80 k
libbrotli x86_64 1.0.9-4.amzn2023.0.2 amazonlinux 315 k
mailcap noarch 2.1.49-3.amzn2023.0.3 amazonlinux 33 k
Installing weak dependencies:
apr-util-openssl x86_64 1.6.3-1.amzn2023.0.2 amazonlinux 15 k
mod_http2 x86_64 2.0.42-1.amzn2023.0.1 amazonlinux 167 k

i-0198e7366c8a95437 (test1)
PublicIPs: 43.204.110.252 PrivateIPs: 172.31.13.231

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28°C Early sunny 25-08-2026
```

```
[ec2-user@ip-172-31-13-231 ~]$ httpd -v
Server version: Apache/2.4.68 (Amazon Linux)
Server built:   Jun  8 2026 00:00:00
[ec2-user@ip-172-31-13-231 ~]$ sudo service httpd start
Redirecting to /bin/systemctl start httpd.service
[ec2-user@ip-172-31-13-231 ~]$ sudo service httpd status
Redirecting to /bin/systemctl status httpd.service
* httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled; preset: disabled)
   Active: active (running) since Tue 2026-08-25 10:26:12 UTC; 19s ago
     Docs: man:httpd.service(8)
    Main PID: 28516 (httpd)
   Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec: 0; Bytes served/sec:  0 B/sec"
     Tasks: 177 (limit: 1059)
    Memory: 13.7M
       CPU: 91ms
    CGroup: /system.slice/httpd.service
            └─28516 /usr/sbin/httpd -DFOREGROUND
              └─28545 /usr/sbin/httpd -DFOREGROUND
                └─28546 /usr/sbin/httpd -DFOREGROUND
                  └─28547 /usr/sbin/httpd -DFOREGROUND
                    └─28592 /usr/sbin/httpd -DFOREGROUND

Aug 25 10:26:12 ip-172-31-13-231.ap-south-1.compute.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
Aug 25 10:26:12 ip-172-31-13-231.ap-south-1.compute.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
Aug 25 10:26:12 ip-172-31-13-231.ap-south-1.compute.internal httpd[28516]: Server configured, listening on: port 80
[ec2-user@ip-172-31-13-231 ~]$ sudo systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
```

i-0198e7366c8a95437 (test1)
PublicIPs: 43.204.110.252 PrivateIPs: 172.31.13.231

Q9. Apache Web Server Configuration

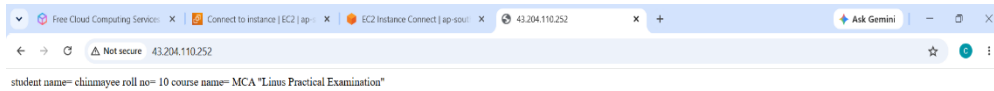
- Find the Apache document root, Create a custom index.html.
- Put the following information on the webpage:
 - Student Name
 - Roll Number
 - Course Name
 - "Linux Practical Examination"
 - Start/restart Apache.
- Access the webpage using the browser or curl. ☐ Display the HTTP response using curl.

Ans:

```
[ec2-user@ip-172-31-13-231 ~]$ cd /var/www/html/
[ec2-user@ip-172-31-13-231 html]$ sudo nano index.html
[ec2-user@ip-172-31-13-231 html]$ sudo nano index.html
[ec2-user@ip-172-31-13-231 html]$ sudo service httpd restart
Redirecting to /bin/systemctl restart httpd.service
[ec2-user@ip-172-31-13-231 html]$ curl http://localhost
student name= chinmayee
roll no= 10
course name= MCA
"Linux Practical Examination"

[ec2-user@ip-172-31-13-231 html]$
```

i-0198e7366c8a95437 (test1)
PublicIPs: 43.204.110.252 PrivateIPs: 172.31.13.231



Section E — Searching, Filtering & Logs

Q11. Search and Text Processing

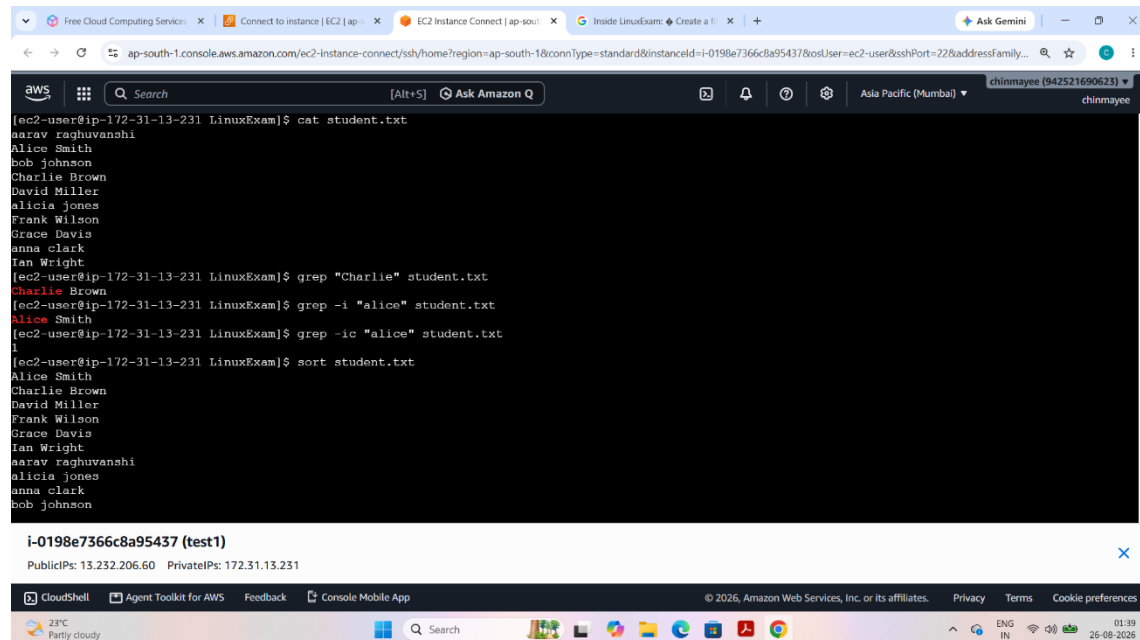
Inside LinuxExam:

- Create a file named `students.txt` containing at least 10 student records.
- Use `grep` to find students matching a specified name or pattern.
- Use `grep` with case-insensitive searching.
- Count matching records, Sort the student records.
- Search for a particular file using `find`.

Find all students whose name contains the letter a.or:

Find all `.txt` files inside the `LinuxExam` directory.

Ans:



Free Cloud Computing Services | Connect to instance | EC2 | ap-south-1 | EC2 Instance Connect | ap-south-1 | Inside LinuxExam: Create a file | Ask Gemini

ap-south-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=ap-south-1&connType=standard&instanceId=i-0198e7366c8a95437&osUser=ec2-user&sshPort=22&addressFamily...

Search [Alt+S] Ask Amazon Q Asia Pacific (Mumbai) chinmayee (942521690623) chinmayee

```
[ec2-user@ip-172-31-13-231 LinuxExam]$ grep -ic "alice" student.txt
1
[ec2-user@ip-172-31-13-231 LinuxExam]$ sort student.txt
Alice Smith
Charlie Brown
David Miller
Frank Wilson
Grace Davis
Ian Wright
aarav raghuvaranahi
alicia jones
anna clark
bob johnson
[ec2-user@ip-172-31-13-231 LinuxExam]$ grep -i "a" student.txt
aarav raghuvaranahi
Alice Smith
Charlie Brown
David Miller
alicia jones
Frank Wilson
Grace Davis
anna clark
Ian Wright
[ec2-user@ip-172-31-13-231 LinuxExam]$ find . -name "*.txt"
./student.txt
./student_backup.txt
./linux_course.txt
[ec2-user@ip-172-31-13-231 LinuxExam]$
```

i-0198e7366c8a95437 (test1)
PublicIPs: 13.232.206.60 PrivateIPs: 172.31.13.231

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